

One can understand this field from chip to ship level only by completing a successful project.

—Sasisekar K., President, RCS Online

second- and third-year engineering course curriculum rather than sector-specific skillsets.”

Solid Verilog/VHDL skills, familiarity with front-end design cycle, knowledge of synthesis and simulation tools, sound digital design fundamentals and knowledge of microprocessors would be an added advantage for any aspiring candidate.

What's the right way to get started?

VLSI circuits are everywhere—your personal computer, your cellphone, your brand new state-of-the-art digital camera or for that matter any electronic gadget you dream to buy. Actually, this field involves packing more and more logic devices into progressively smaller areas. So the circuits that would have taken board-full of space can now be put into a small space few millimetres across!

According to Vivek Pawar, CEO, Sankalp Semiconductor, “The best way to start work on VLSI is to build core competency in this field. Component-level knowledge of electronic circuits and devices and thorough understanding of network theory, linear and digital design concepts are must-have.” Further, he emphasises on step-by-step knowledge acquisition of digital design, analogue design, board design and system design using different embedded tools.

To achieve this goal, Sankalp picks up second- and third-year electronics engineering students under its industry-academia ecosystem ‘Eklakshya’ and trains them following the ‘learning by doing’ principle.

According to Dr P.S. Bhat, director, Eklakshya, “You may start with a typical analogue or digital design flow. For example, a digital design

flow is specification→architecture→RTL coding→RTL verification→synthesis→backend→tape out to foundry to get the end product.” Here the end product is a wafer with repeated number of identical ICs. All modern digital designs start with a designer writing a hardware description of the IC (using HDL) in Verilog/VHDL. A Verilog or VHDL program essentially describes the hardware (logic gates, flip-flops, counters, etc), the functionality and circuit block interconnects. Various electronic design automation (EDA) tools are available to synthesise a circuit based on the HDL. The most widely used synthesis tools come from Synopsys and Cadence.

Without going into details, you may consider VHDL as ‘C’ language of the VLSI industry. VHDL stands for ‘VHSIC hardware definition language.’ This language is used to design the circuits at a high level, in two ways. It can either be a behavioural description of what the circuit is supposed to do or a structural description of what the circuit is made of. There are other languages too for describing circuits, such as Verilog, that work in a similar fashion. Both forms of description are then used to generate a very low-level description that actually spells out how all this is to be fabricated on the silicon chips. This results in the manufactur-

ing of the intended IC.

For an analogue design, the flow may vary to some extent: specifications→architecture→circuit design→simulation→layout→parametric extraction/back annotation→final design→tape out to foundry.

While digital design is highly automated now, very small portion of analogue design can be automated. For analogue, there is a hardware description language called AHDL, but it is not widely used as it does not accurately give the behavioural model of the circuit because of the complex analogue behaviour of the circuit. Many analogue chips are termed as ‘flat’ or non-hierarchical designs. This is true for small-transistor-count chips such as operational amplifiers, filters or power management chips. For more complex analogue chips such as data converters, the design is done at the transistor level, building up to the cell level, then the block level and finally integrated at the chip level. Not many EDA tools are available for analogue design even today and thus analogue design remains a difficult art. SPICE is the most useful simulation tool for analogue as well as digital design.

Keep in mind that digital VLSI circuits are predominantly CMOS-based. So the way normal blocks like latches and gates are implemented is different from what you have seen so far. However, the behaviour remains the same. All the miniaturisation involves new things to consider. A lot of thought has to go into the actual implementation as well as design.

Further, Dr Bhat points out, “Lay-

Laying out the circuit components is a task common to all branches of electronics. The speciality in VLSI layout is the number of possible ways available to do this.



—Dr P.S. Bhat, Director, Eklakshya